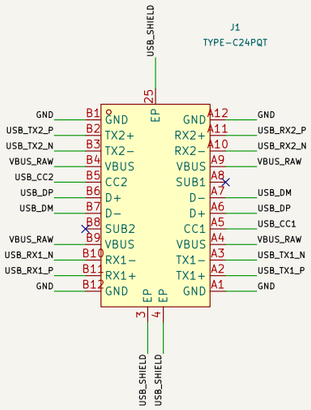
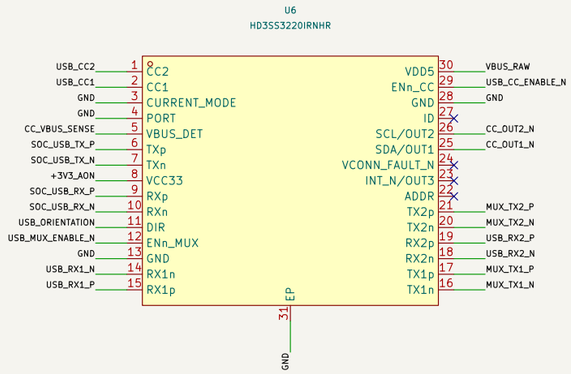
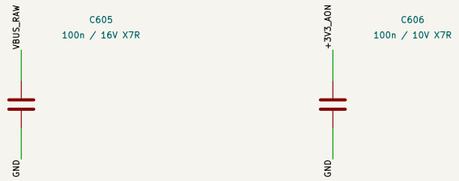
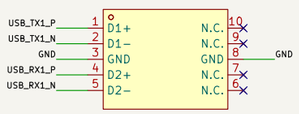
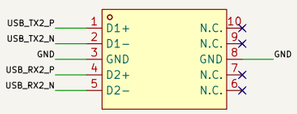
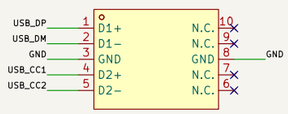


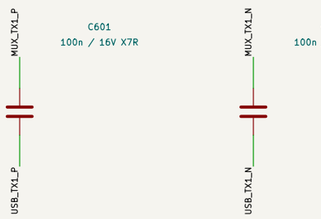


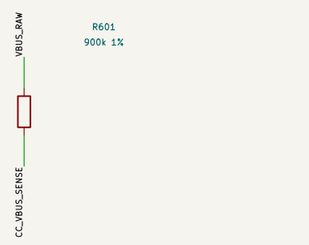












USB-C DATA FRONTEND — ENGINEERING DRAFT, NOT FOR FABRICATION

Sink/UFP only, GPIO mode: PORT=GND, ADDR=NC. No USB Power Delivery negotiation.

Power circuitry is NOT included: VBUS protection, inrush, +3V3_AON and load gating remain required.

Require CC_OUT1_N=0 AND CC_OUT2_N=0 for the proposed 3 A startup policy; OUT2 alone is insufficient.

USB_CC_ENABLE_N requires verified VDD5/VCC33 sequencing (TI section 6.3.12, >=2 ms when controlled).

USB_MUX_ENABLE_N must default disabled until main SoC supplies are valid.

TX AC coupling is on the connector side of the mux, following TI UFP reference. No duplicate SoC TX capacitors.

USB_SHIELD termination and connector footprint remain to validate. Passive MPNs are not selected.

RK3568 endpoints: T28/T27 TX+/−; R28/R27 RX+/−; P27/P28 DP/DM; M24 VBUSDET needs its own sensing circuit.

Sources: TI HD3SS3220 SLLSES1E, TPD4EUSB30 SLVSAC2G; Rockchip Hardware Design Guide V1.2.

Incomplete system — not for manufacture

Sheet: /
File: usb-c-frontend.kicad_sch

Title: Phoenix G1 — USB-C data frontend

Size: A3	Date:	**Rev: DRAFT**
KiCad E.D.A. 10.0.7		Id: 1/1